Precambrian Research 186 (2011) 193–204

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Paleomagnetic constraints on an Archean–Paleoproterozoic Superior–Karelia connection: New evidence from Archean Karelia

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ARTICLE INFO

Article history:
Received 29 June 2010
Received in revised form 20 January 2011
Accepted 27 January 2011
Available online 2 February 2011

Keywords:
Archean
Paleoproterozoic
Baltica
Karelian craton
Paleomagnetism
Palaeogeography

ABSTRACT

Charno–enderbitic granitoids in the Karelia craton of the Fennoscandian shield have been studied paleomagnetically. The characteristic remanence component has a steep negative inclination and is interpreted to record magnetization at a maximum age of 2684 ± 2 Ma. Consistently stable results were obtained from 12 sites in the Koitere area, corresponding to regions with high positive magnetic anomalies and high remanence intensities. Petrographic studies, coupled with rock magnetic investigations, indicate that the remanence resides in fine SD/PSD magnetite grains formed during Neoarchean clinopyroxene alteration. Cross-cutting vertical/subvertical Paleoproterozoic dolerite dykes suggest that the Koitere granitoids are in their original orientations and were not affected by Svecofennian deformation at ca. 1.9–1.8 Ga.

The Koitere granitoids have an opposite polarity compared to the steep positive inclination remanence direction of the previously studied ca. 2.63 Ga Varpaisjärvi enderbites and granulites. The data from Koitere and Varpaisjärvi imply that at ca. 2.7–2.6 Ga the Karelia craton was located at high latitudes of 80–60°, whereas previous paleomagnetic data from ca. 2.5 Ga formations in the Vodlozero terrane in NW Russia indicate a near-equatorial position.

Comparison of paleomagnetic data from the Koitere and Varpaisjärvi granulite-grade rocks with rocks of similar age in the Superior craton shows that at ca. 2.7–2.6 Ga the Superior and Karelia cratons were located at high latitudes and in close proximity, although the present data cannot demonstrate that the cratons were amalgamated. However, during the Archean–Paleoproterozoic transition at ca. 2.50 Ga both cratons experienced significant rotation and drifting to near-equatorial paleolatitudes, suggesting that the Superior and Karelia cratons may have been attached at that time.

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1. Introduction

Paleomagnetic data can provide an important record of plate configurations and cratonic evolution, and studies on Archean cratons have been carried out over several decades. Recent studies on the Kaapvaal and Pilbara cratons have significantly increased the Archean dataset in recent years years (de Kock et al., 2009; Strik et al., 2003, 2007), although reliable Archean data are still sparse. Investigations from the Karelia craton of the Fennoscandian shield (Baltica) and from the Superior craton of Laurentia have provided relatively good-quality paleomagnetic data, but in most cases the data are ambiguous due to uncertainty in the timing of the remanent magnetization and demonstration of its primary origin. A recent Superia supercraton model (Bleeker and Ernst, 2006; Bleeker et al., 2008) suggests a tight link between the Superior and Karelia cratons from the Neoarchean to Paleoproterozoic at ca. 2.6–2.0 Ga, based mainly on correlations between similar-aged Paleoproterozoic dyke swarms on both cratons. However, the model does not consider paleomagnetic data, likely due to poor data or age discrepancies between the cratons. The model suggests that prior to the rifting and emplacement of voluminous layered intrusions and dyke swarms at ca. 2.5–2.4 Ga, a unified Neoarchean Superia craton existed. This hypothesis is also supported by synchronous Neoarchean magmatic events and high-grade metamorphism at ca. 2.66–2.63 Ga in both cratons. In order to provide supplementary paleomagnetic constraints for the existence of a Neoarchean supercraton we present new paleomagnetic data from well-preserved Archean rocks in eastern Finland (see Mertanen and Korhonen, 2008).

In the Fennoscandian shield, existing Archean paleomagnetic data has been acquired from the Karelia and Kola cratons in Finland and NW Russia. In most cases the results are ambiguous because of polyphase tectonic movements, multicomponent remanent magnetizations and pervasive overprinting during the Paleoproterozoic. Obtaining unambiguous Archean data is chal-

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doi:10.1016/j.precamres.2011.01.018

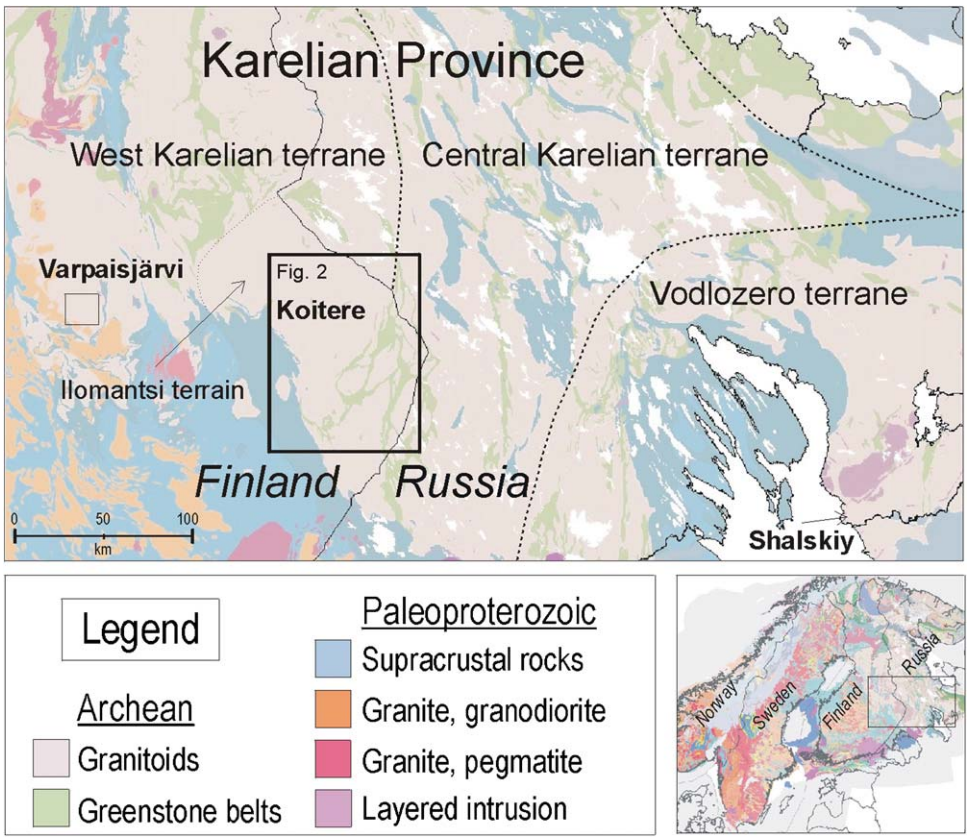


Fig. 1. Generalized geological map of the Karelia craton in the eastern Fennoscandian shield (after Koistinen et al., 2001). West Karelian and Central Karelian Terranes and the Vodlozero Terrane (after Hölttä et al., 2008) are outlined with a thicker dotted line and the Ilomantsi terrain within the West Karelian Terrane with a thinner dotted line. The study area of Koitere is shown by a rectangle. Previous paleomagnetic study areas of Varpaisjärvi and Shalskiy are indicated.

lenging because demonstrating the primary nature of an Archean remanence is difficult by standard field tests and the accurate dating of the remanence components due to overprinting is not straightforward. This study aims to explore the most reliable Archean paleomagnetic data of Baltica and presents new data from the Archean (ca. 2.73 Ga) Koitere charno–enderbites in the southern Karelia craton of eastern Finland (Fig. 1). The data from the Koitere granitoids are compared with data from two other Neoarchean rocks (Fig. 1): the ca. 2.63 Ga Varpaisjärvi granulites in central Finland (Mertanen et al., 2006a) and the ca. 2.5 Ga Shalskiy gabbronorite dyke in the Vodlozero terrane in northwestern Russia (Mertanen et al., 2006b; Bleeker et al., 2008). These three rock units have been shown to preserve Archean information on the basis of isotopic dating, petrographic studies, and the absence of pervasive ca. 1.9–1.8 Ga Svecofennian overprinting. The Karelian paleomagnetic data are compared with similar age data from the Superior craton of Laurentia.

2. Geological setting of the Koitere area

The Koitere granitoid area is located in the Ilomantsi terrain of the Western Karelian terrane of the Karelia craton where the Archean basement rocks consist of TTG gneisses and migmatites, granite–greenstone belts and granitoids (Hölttä et al., 2008; Halla et al., 2009; Heilimo et al., 2010a). The Archean bedrock is generally metamorphosed to amphibolite-facies, except certain areas that preserve granulite-facies metamorphic assemblages. The presently studied granodiorites in the Koitere area are located within such a granulite-grade area. The area is characterized by a pronounced high magnetic anomaly approximately 200 km^2 in size (Fig. 2), caused by well-preserved pyroxene + magnetite-bearing granodiorites (Korhonen et al., 2006). The intervening shear zones and retrograde gneisses (Halla, 2002) have lower magnetic intensities. According to Halla (2005) the charno–enderbites represent high Mg/Fe granitoids known as sanukitoids, and the overall chemical composition of the Koitere sanukitoids is silimar to Archean sanukitoid suites in the western Superior craton (Halla, 2005).

In the Fennoscandian shield the sanukitoid-type rocks were emplaced in the Karelia craton at ca. 2.74–2.70 Ga (Bibikova et al., 2005; Halla, 2002, 2005; Halla et al., 2009; Heilimo et al., 2007; Käpyaho, 2007) and are younger than the surrounding ca. 2.83–2.75 Ga TTG gneisses (e.g. Käpyaho, 2007; Käpyaho et al., 2007). The sanukitoids occur in two zones covering areas both in eastern Finland and in NW Russia (e.g. Lobach-Zhuchenko et al., 2000, 2005; Bibikova et al., 2005). In the eastern zone, in the Vodlozero terrane in Russia, the sanukitoids are slightly older (ca. 2.75–2.73 Ga) than in the western zone, in the West Karelian terrane (ca. 2.72–2.70 Ga). The relationship between the sanukitoids and granulite-facies metamorphism is still unclear. In the Koitere area, there is no evidence for a time gap between the crystallization of the granitoids and granulite-facies metamorphism (Halla, 2002, 2005), therefore the formation of sanukitoids (charno–enderbites) may be a result of either crystallization under high-grade conditions or high-grade metamorphism. U–Pb zircon geochronology using conventional methods from the Koitere sanukitoids suggests crystallization at 2733 ± 29 Ma (Halla, 2002). Recent dating using secondary ion mass spectrometry (SIMS) methods from one of the sanukitoids studied here yields a zircon age of 2722 ± 6 Ma (Heilimo et al., 2011).

The youngest $^{207}Pb/^{206}Pb$ zircon ages from leucosomes in migmatites in the Koitere area are between ca. 2.71 Ga and

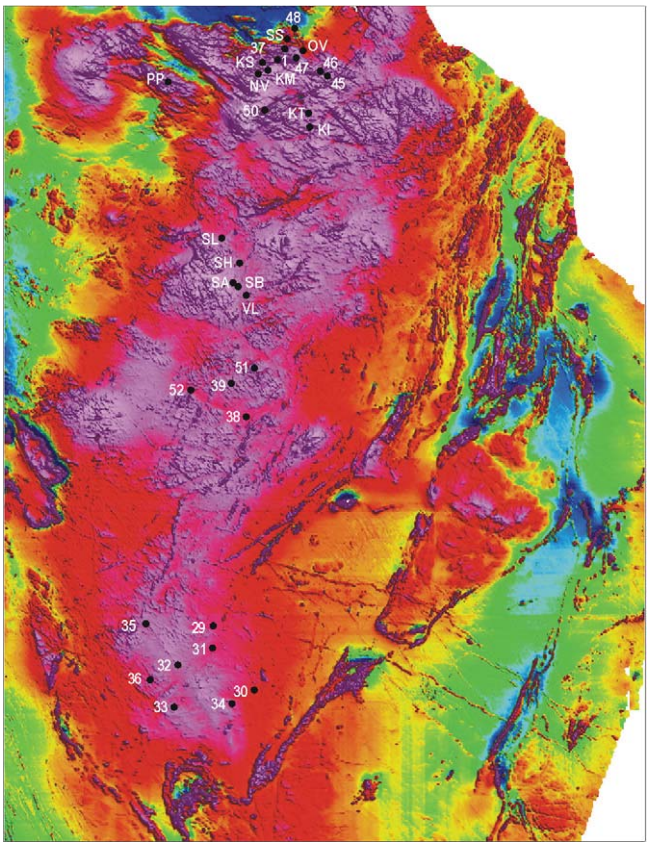


Fig. 2. Aeromagnetic map of the study area and sampling sites. Stable paleomagnetic results were obtained from the northern sites that show the highest magnetic intensities represented by red to purple shading, and sporadically in the southern sites. Base map compiled by Meri-Liisa Airo, GTK.

2.68 Ga, and this age range is attributed to prolonged migmatization (Käpyaho et al., 2007). These ages are in agreement with a $^{207}Pb/^{206}Pb$ monazite age of 2684 ± 2 Ma from a well-preserved charno–enderbite in the Koitere area (Halla, 2002), and provide the best age estimate for granulite-facies metamorphism. However, because monazite can preserve isotopic ages associated with crystallization at temperatures higher than the magnetite unblocking temperature of ca. 580 °C (e.g. see Kelsey et al., 2008) and as the remanent magnetization records cooling after the final high-temperature event, this age gives the maximum estimate for the timing of remanence acquisition. In addition, according to Käpyaho (2007 and references therein) the final tectonothermal events related to the cratonisation of the Archean crust in eastern Finland took place at ca. 2.72–2.63 Ga, providing a minimum age estimate. Based on these constraints, the timing of remanence acquisition in the Koitere granitoids is inferred to have occurred between ca. 2.68 and 2.63 Ga.

3. Paleomagnetic studies

3.1. Field investigations

The sampling region in Koitere covers an area of about 40 km × 100 km, and 174 samples were taken from 32 sites (Fig. 2). Sampling was done both with mini-drills and as hand samples and oriented with sun and magnetic compasses. Most of the samples were taken from well-preserved and undeformed two-pyroxene sanukitoids that represent granulite-facies granodiorites. The sanukitoids are surrounded by retrogressed TTG gneisses and shear zones that were affected by Proterozoic events. Some samples were taken from these zones for comparison.

3.2. Laboratory methods

Standard cylinders with a diameter of 2.5 cm and length of 2.2 cm were prepared in the laboratory. Density and magnetic susceptibility were measured from most specimens prior to paleomagnetic measurements. Natural remanent magnetization (NRM) intensities and Koenigsberger ratios (Q-value) were measured from those specimens that were also demagnetized. For paleomagnetic studies, the specimens were either demagnetized in an alternating field (AF) in 12–15 incremental steps up to 100 or 160 mT or thermally, up to a maximum temperature of 620 °C. Remanent magnetization was measured with a cryogenic three-axis Squid (RF)-magnetometer after each demagnetization step. Separation of remanent components was done with principal component analysis (Kirschvink, 1980) using the Tubefind program (Leino, 1991). Fitting of lines (Zijderveld, 1967), with a minimum of three demagnetization points, was done with the maximum angular deviation of the line up to 6° or as high as 10° in some cases. Site directions were calculated according to Fisher (1953). Magnetic carriers were identified by thermomagnetic analysis on eight samples using an Agico CS2-KLY2 Kappabridge, applying stepwise heating up to 700 °C. For characterizing the magnetic domain sizes and thus the ability of rocks to preserve ancient remanences, modified Lowrie–Fuller tests (Lowrie and Fuller, 1971) were performed on five specimens. The samples were first stepwise demagnetized in alternating fields up to 100 or 160 mT, and then given an anhysteretic remanent magnetization (ARM) in a peak AF of 100 mT and a biasing field of 50 μT. The ARM was then progressively AF demagnetized in peak fields up to 100 or 160 mT. The specimens were then pulse magnetized in a Molspin pulse magnetizer in 1.5 T to produce a saturation isothermal remanent magnetization (SIRM) after which they were progressively AF demagnetized in peak fields up to 100 or 160 mT. For petrographic and microprobe analysis a total of 35 polished thin sections were studied. The GMAP program was used for presenting the paleomagnetic poles and for calculating the positions of the cratons.

4. Results

4.1. Magnetic mineralogy

Site mean petrophysical data for those samples that yield consistent stable paleomagnetic results are shown in Table 1. Total (NRM) magnetizations are high, the remanence intensities being typically thousands of mA/m and susceptibilities tens of thousands ($\times 10^{-6}$) of SI units. Koenigsberger ratios range from 0.2 to 5.5, with most values between 1 and 2. In general, the stable paleomagnetic results were obtained from samples with the highest remanence intensities and Koenigsberger ratios. The relatively low Koenigsberger ratios, although signifying high remanent magnetization compared to the induced magnetization, indicate that the rocks were not affected by lightning, which typically produces isothermal remanent magnetizations (IRMs) with much higher ratios.

Thermomagnetic analyses (Fig. 3) show reversible heating and cooling curves with a Curie point of ca. 580 °C, indicative of pure magnetite. The magnetization is unblocked within a very narrow temperature range of 560–580 °C suggesting fine-grained magnetite as the remanence carrier. A small Hopkinson peak in some samples further implies single domain (SD) or pseudo-single domain (PSD) magnetic domain size. This observation is confirmed with modified Lowrie–Fuller tests (Fig. 4) where the ARM curves show harder coercivities compared to the IRM curves, consistent with SD or PSD particles (Lowrie and Fuller, 1971; Dunlop, 1983).

Table 1
Site mean paleomagnetic and petrophysical results, Koitere (Glat = 63.3°N, Glong = 30.6°E).

Site	B/N/n	Dec (°)	Inc (°)	95 (°)	*k*	*J* (mA/m)	Susc. $\times 10^{-6}$	Q	Dens.	*x*	*y*
KI	1/[a]2	0.4	−75.1	–	–	3664	17436	5.3	2670	7014250	4534900
KM	1/[a]2	293.2	−82.0	–	–	213	25177	0.2	2703	7020800	4530100
KS	[a]6/11	175.9	−76.3	3.9	304.2	5886	82348	1.9	2799	7021130	4529700
NV	[a]7/10	161.3	−83.9	5.8	109.0	4190	78471	1.5	2822	7020310	4529300
OV	[a]5/7	143.0	−75.6	14.0	30.7	1210	30706	1.0	2681	7022290	4533750
SS	[a]6/11	26.3	−75.3	6.9	95.7	2300	60790	1.1	2752	7024320	4532480
01	[a]2/2	154.0	−73.0	–	–	700	41290	0.4	2747	7021820	4531040
45	1/[a]2	337.5	−79.8	–	–	–	–	–	–	7019760	4536440
46	[a]4/6	331.1	−89.0	21.7	18.9	–	–	–	–	7020140	4535720
47	[a]2/2	181.2	−78.2	–	–	1750	68000	0.7	2752	7022170	4533030
48	1/1	25.1	−72.9	–	–	840	2960	7.1	2977	7025010	4533270
50	[a]2/2	267.8	−52.2	–	–	–	–	–	–	7016160	4529840
Mean	[a]12/38/58	260.5	−88.7	9.2	23.2						
Mean	[a]5/28/45	133.9	−85.1	10.9	49.9						

Mean paleomagnetic pole: *N* = 12: −62.6°N, 213.1°E, dp/dm = 18.4°/18.4°, A95 = 15.9°, *K* = 8.4. *N* = 5: −67.5°N, 192.6°E, dp/dm = 21.4°/21.7°, A95 = 19.5°, *K* = 16.4. Note. B/N/n, number of sites/samples/specimens used for mean calculations.

[a] Denotes the statistical level used for mean calculation; mean values calculated for *N* = 12 (all sites) and *N* = 5 (sites with $N \geq 5$ samples); Dec and Inc, the mean declination and inclination, respectively; 95, radius of the circle of 95% confidence; *k*, Fisher (1953) precision parameter; *J*, remanence intensity; Susc., magnetic susceptibility; *Q*, Koenigsberger ratio; Dens., density. *x*, *y* give the coordinates of sampling sites (UTM coordinate system). Mean paleomagnetic pole: Plat (°N), Plong (°E); dp/dm, semi-axes of the 95% confidence around the pole; A95, radius of the circle of 95% confidence around the pole; *K*, Fisher precision parameter of the mean pole.

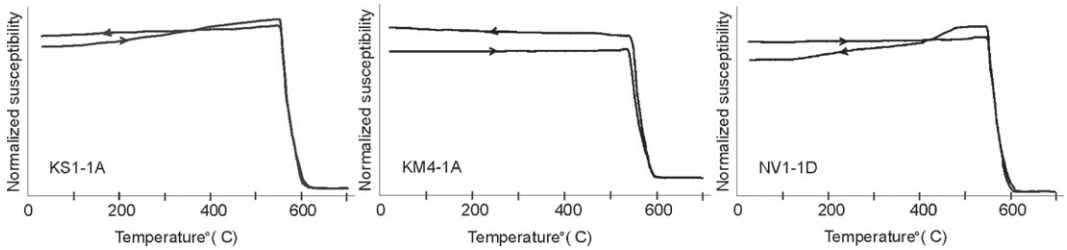


Fig. 3. Thermomagnetic analyses from specimens of sites KS, KM and NV that carry the stable ChRM. Based on a Curie temperature of 580 °C, the remanent magnetization resides in magnetite.

The Koitere granitoids were studied with electron microscope and optical microscope in order to identify the carriers of the stable remanent magnetization and possible mineralogical reactions related to their formation. Petrographic studies show that the stable paleomagnetic results are commonly obtained from those sanukitoid samples that contain fine-grained magnetite that was formed during hydrous alteration of clinopyroxene (Fig. 5). As will be discussed later, this alteration is interpreted to be Archean in age.

4.2. Paleomagnetism

A stable, internally consistent characteristic remanent magnetization component (ChRM) was isolated in the northern part of the study area that shows the strongest magnetic intensities (Fig. 2). In the southern sites the ChRM component was isolated only sporadically. Table 1 lists the mean remanence directions of the sites. The mean direction and pole position were calculated for each site (*N* = 12), including sites with only one sample, and for five sites where the ChRM was isolated in a minimum of four samples. Examples of demagnetization behaviours are shown in Figs. 6 and 7, and the mean directions of five sites and the overall mean are shown in Fig. 8.

The ChRM has a consistent steep (almost vertical) upward-pointing direction, but due to these steep inclinations the declinations vary greatly. AF demagnetization results (Fig. 6) imply that the ChRM is carried by a high coercivity mineral, consistent with PSD/SD-sized magnetite. According to thermal demagnetizations (Fig. 7), the maximum unblocking temperatures are close to 580 °C, consistent with magnetite as the remanence carrier. In low temperatures and AF fields, a small steep downward component was isolated in most samples. It is likely of viscous origin as its direction is close to the present Earth's magnetic field (PEF, Figs. 6 and 7) in the area.

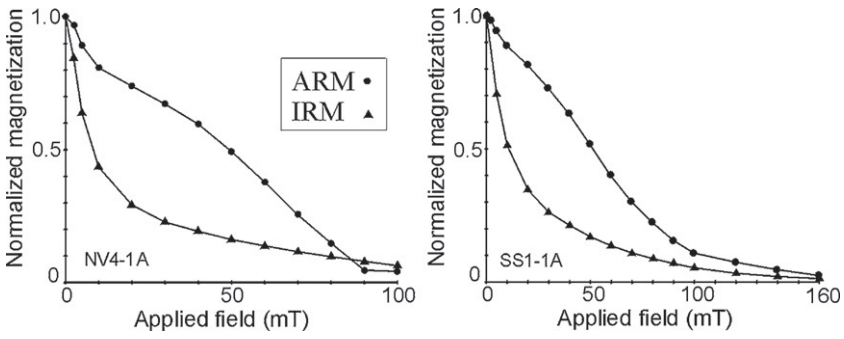


Fig. 4. Modified Lowrie–Fuller tests for samples from sites NV and SS that carry the ChRM. Circles denote AF demagnetization of ARM and triangles of SIRM. The higher coercivities of ARM indicate that the remanence is carried by SD/PSD fractions.

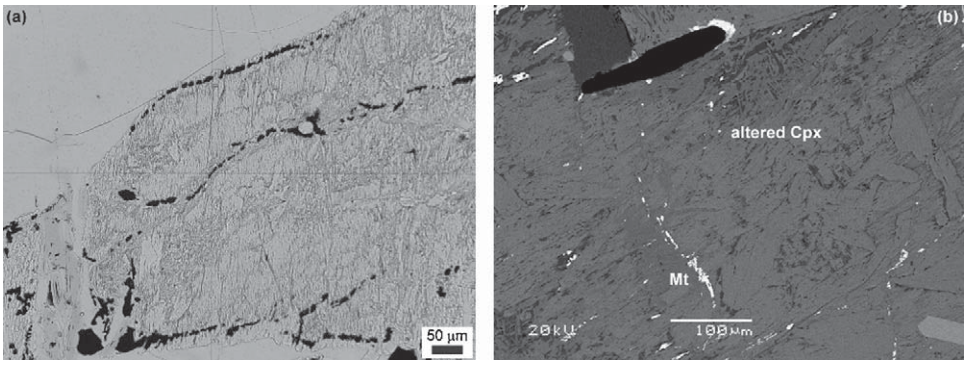


Fig. 5. (a) Photomicrograph showing PSD magnetite (opaque) from sample 47.4, (b) Backscatter electron image of magnetite (Mt) and of pseudomorph after clinopyroxene (Cpx) from sample 47.4.

In some samples, especially from the southern and northwestern sites, a secondary downwards NW-pointing moderate inclination component was isolated (see Fig. 9). It occurs characteristically in those samples that have coarser magnetic grain size and which have thus been vulnerable to subsequent remagnetization. The samples in the northern sites that carry the ChRM and highest magnetization intensities typically do not carry this superimposed component. In many cases the secondary NW-pointing component is unstable and could not be isolated by principal component analyses, and the direction is evident only on stereoplots. It is regarded as a partial remagnetization acquired during the Paleoproterozoic Svecofennian orogeny (see e.g. Krasnova and Gooskova, 1995; Mertanen, 1995; Mertanen et al., 1999). The component occurs only sporadically in the whole area and its occurrence may be related to hydrous alteration that is observed petrographically (Korhonen et al., 2006). In some samples it was isolated in a temperature range of 200–400 °C, but higher temperatures up to 580 °C and in one case up to 680 °C were also observed. In those rare cases where this component occurs together with the steep upward-pointing ChRM, it is isolated at lower temperatures and AF fields, consistent with it being a later overprint (Fig. 9). Thermomagnetic analyses show that the remanence is carried by magnetite or titanomagnetite.

In some samples an E–SE-pointing moderate inclination remanence component was isolated. In most cases it does not occur with the ChRM, except in two samples in the northern area (sites 46 and 47). The direction is close to the remanence that is characteristic of the ca. 2.45 Ga dykes and layered intrusions in Fennoscandia (e.g. Mertanen et al., 1989, 1999, 2006b), and may reflect partial reheating of the crust at that time (see also Vaasjoki et al., 1999). As discussed in Mertanen et al. (2006b), it is probable that the emplacement of layered intrusions and associated dykes which are thought to be related to a regional-scale mantle plume event, may be responsible for this partial remagnetization.

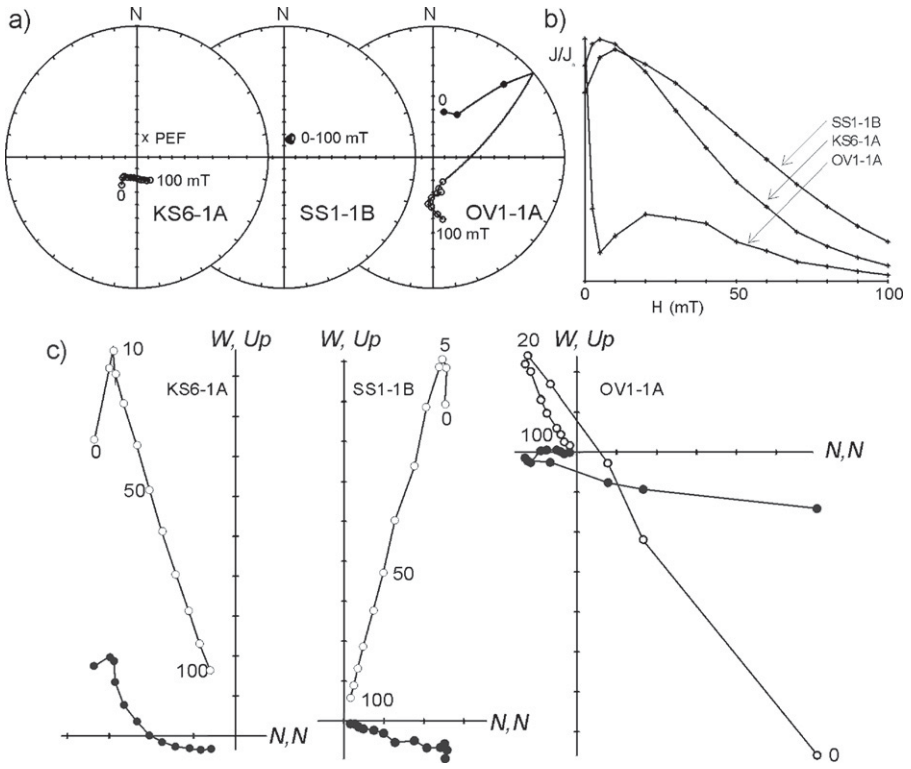


Fig. 6. Examples of AF demagnetization behaviour of samples from sites KS, SS and OV, (a) stereoplots, (b) intensity decay curves of AF demagnetization, (c) orthogonal vector plots where open (closed) symbols denote vertical (horizontal) planes. Numbers refer to AF fields (mT). PEF shows the present Earth's field direction at the sampling location.

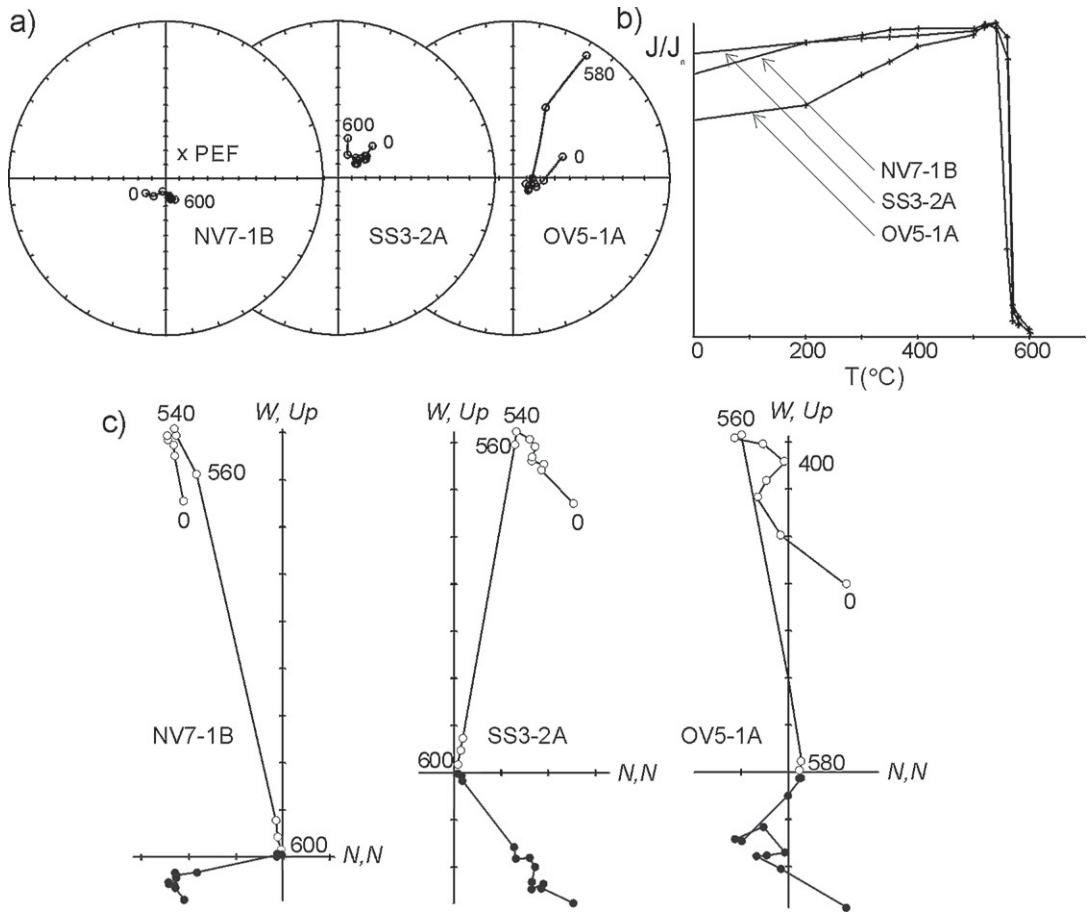


Fig. 7. Examples of thermal demagnetization behaviour of samples from sites NV, SS and OV. Numbers on vector plots refer to temperatures (°C). For other explanations, see Fig. 6.

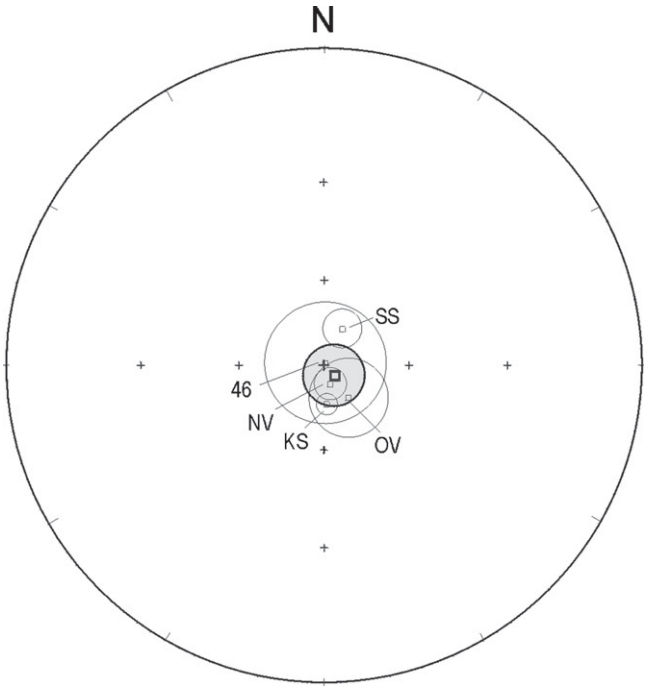


Fig. 8. Mean remanence directions of five sites with the mean cones of α95. The overall mean direction with the α95% confidence cone is shaded. Open symbol denotes negative inclination. Other ChRM directions with less than 5 samples per site are listed in Table 1.

5. Discussion

5.1. Origin and age of the ChRM in Koitere

A stable consistent ChRM component was isolated from granitoids in 12 sites that consist mainly of high-grade metamorphic charno–enderbites and monzonites. The stable results were only obtained within the region characterized by a high positive magnetic anomaly in the northern area (Fig. 2). The high magnetic anomalies reflect the magnetic susceptibilities carried by coarse-grained igneous magnetite. The areas of relative magnetic lows are dominated by more felsic lithologies that did not favor the formation of magnetite. The areas of high magnetic susceptibility are also areas of high remanence intensity. Based on rock magnetic tests, the high remanence intensities are carried by fine-grained magnetite. It is implied that this magnetite was formed during clinopyroxene alteration and crystallization of secondary fine-grained SD/PSD magnetite along fractures (Fig. 5). Electron microprobe and thin section observations are consistent with rock magnetic studies which show that the stable remanence is carried by high-coercivity and high-unblocking temperature SD/PSD magnetite. In that sense the remanence of the magnetite may represent a chemical or thermochemical remanent magnetization (CRM/TCRM). Geological studies from central Finland demonstrate that a major tectonothermal event took place at ca. 2.7–2.6 Ga when Archean crust was metamorphosed and migmatized (Käpyaho, 2007 and references therein) under fluid-present conditions (Nehring, 2007). This widespread metamorphic event and associated fluid flow also affected the Archean rocks in the Koitere area; therefore, the alter-

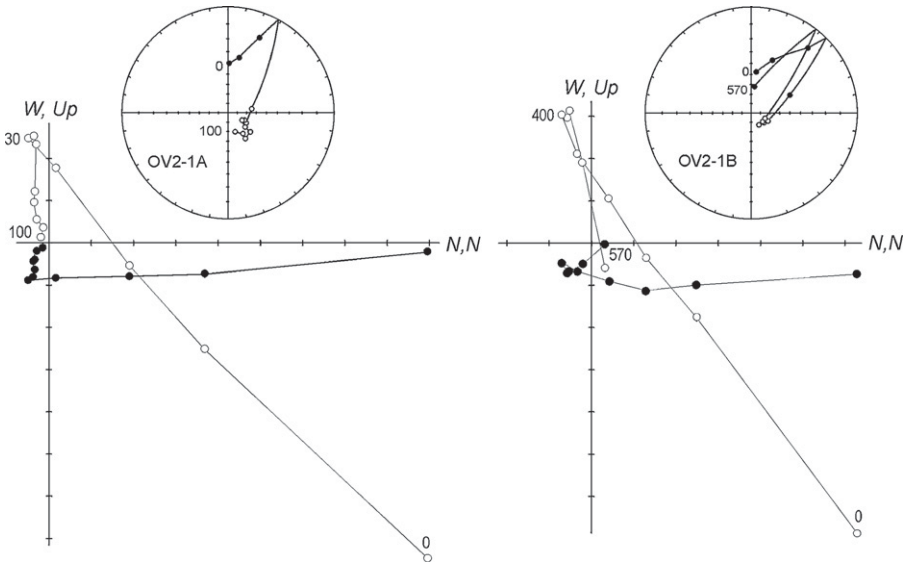


Fig. 9. AF and thermal demagnetization behaviour of sample OV2 showing a typical NW-directed, downward-pointing secondary Svecofennian overprint isolated in AF fields of 0–30 mT and temperatures of 0–400 °C. For explanations, see Figs. 6 and 7.

ation of clinopyroxene in these rocks was likely a late Archean process. The $^{207}Pb/^{206}Pb$ monazite age of 2684 ± 2 Ma (Halla, 2002) obtained from the same location as one of the sites with stable ChRM (KI, Table 1) constrains the maximum timing of the remanent magnetization.

Demonstrating the Archean origin of the ChRM is difficult as there are no suitable field tests for Archean granitoids. Our interpretation of the Archean origin is consistent with the following: (1) The stable and internally consistent ChRM is carried by SD/PSD magnetite grains which are able to preserve the ancient remanence. (2) The ChRM was obtained only from the pyroxene + magnetite-bearing high-grade granitoids, whereas the Archean TTG gneisses and sheared granitoids surrounding the sanukitoids do not carry the ChRM component. These latter rocks are associated with lower magnetic anomalies, typically have coarser grain size, and carry either a recent viscous remanent magnetization or the downwards NW-directed overprint. On the basis of the remanence direction that is well established from previous studies (e.g. Mertanen et al., 1999), the overprint was likely acquired during the Svecofennian orogeny at ca. 1.88–1.84 Ga. Although the Svecofennian orogeny was a pervasive event throughout the area, it was not able to remagnetize those high-grade rocks that carry the high coercivity and high unblocking temperature magnetite. In those few cases where both the Svecofennian and the ChRM component occur together, the Svecofennian component is revealed at lower temperatures, implying a maximum overprint temperature of about 500 °C. The Svecofennian remagnetization most likely represents a thermochemical/chemical remanent magnetization corresponding to lower temperatures, whereas the ChRM component has high unblocking temperatures (570–600 °C) and was not affected by the lower temperature metamorphism. (3) The occurrence of vertical/subvertical Paleoproterozoic dolerite dykes with ages ranging from ca. 2.5 Ga to 1.95 Ga are abundant in the eastern Finland Archean complex (Kontinen and Paavola, 2006, and references therein). Although the dykes were metamorphosed under amphibolite-facies conditions (Kontinen and Paavola, 2006), they are relatively undeformed. Their structural relationship with the host granitoids suggests that the granitoids record paleohorizontal orientations and that the blocks were not significantly deformed during events associated with the Svecofennian orogeny at ca. 1.9–1.8 Ga.

These observations indicate that a Neoarchean remanence has been preserved in the Koitere charno-enderbites. Similar remanence directions with steep negative inclinations have only been observed in two other studies of Precambrian rocks in Fennoscandia. A steep upward-pointing remanence was isolated in a boninitic Viianki dyke with a minimum age of 2390 Ma (Vuollo and Huhma, 2005). In the Viianki dyke the steep negative component is the characteristic remanence of the dyke with $D = 173.9°$, $I = -72.3°$ ($a95 = 17.6°$, $k = 50.2$, 3 sites/6 samples, S. Mertanen, unpubl. data), which differs significantly from the typical remanence direction of similar-aged rocks (e.g. Mertanen et al., 1999). The origin of the negative inclination component has not been resolved in the Viianki dyke. Similarly, the Kautokeino dykes in northern Norway (Sm–Nd age (whole rock, clinopyroxene, plagioclase): 1066 ± 34 Ma) carry a relatively steep upward-pointing direction ($D = 315.1$, $I = -68.4$, Mertanen et al., 1996). However, this age does not correspond to any known geologic events or other isotopic data to support resetting at that time, therefore it is highly unlikely that the remanence of the Koitere granitoids would be of that age.

5.2. Other Archean formations of Fennoscandia

The most compelling previous Archean paleomagnetic data from the Karelia craton have been obtained from granulite-grade enderbites in the Varpaisjärvi area in the Iisalmi terrain in central Finland (Neuvonen et al., 1981, 1997; Mertanen et al., 2006a) and from the Shalskiy gabbronorite dyke and the granulite-grade gneisses in the Vodlozero terrane in NW Russia (Krasnova and Gooskova, 1990; Mertanen et al., 2006b) (Fig. 1). The Koitere (e.g. Halla, 2002, 2005; Halla et al., 2009; Heilimo et al., 2010a; Kontinen et al., 2007), Varpaisjärvi (e.g. Hölttä et al., 2000), and Vodlozero (e.g. Hölttä et al., 2008; Slabunov et al., 2006) areas show various similarities. In all sampling areas the granitoids were metamorphosed under granulite-facies conditions. They are generally well-preserved and show little evidence of Svecofennian overprinting, although the surrounding or lower-grade Archean rocks and cross-cutting Early Paleoproterozoic dykes have been partially or totally remagnetized during this event. Furthermore, the granitoids show high magnetic anomalies whereas the surrounding TTG gneisses are weakly magnetized. The high-grade rocks have high remanence stability, and the ChRM differs from known Pro-

Table 2
Paleomagnetic poles from the Karelia and Superior cratons.

Formation	B/N/n	Plat (°N)	Plong (°E)	A95	Age of ChRM (Ma)	Ref.
Karelia						
ca. 2680 Ma						
Koitere sanukitoid	[a]5/28	−67.5	192.6	19.5	2680–2630	1,2,3
ca. 2600 Ma						
Varpaisjärvi granulite	[a]14/65	−67.4	154.1	10.0	2630–2590	4,5
Varpaisjärvi granulite	8/12	−63.9	133.0	7.9	2630–2590	6,5
ca. 2500 Ma						
Shalskiy dyke	1/[a]5(16)	22.7	222.1	10.7	2510.6 ± 1.5	7,8
Shalskiy dyke	3/[a]18	24.0	224.0	5.3	2510.6 ± 1.5	9,8
Vodla River gneisses	[a]20/30	18.0	212.0	7.1	2540 ± 50	9,10
Superior						
ca. 2680 Ma						
Otto Stock lamprophyre	/[a]71	69.0	227.0	5.0	<2680 ± 1	13,14
ca. 2600 Ma						
Pikwitonei granulite (rot.)	/[a]30	27.9	217.9	6.5	2637 ± 2	15,16
ca. 2500 Ma						
Ptarmigan-Mistassini dykes	[a]3/	−45.3	213.0	13.8	2515–2505	17,18,19,20

Note: B/N/n, number of sites/samples/specimens.

[a] Selected statistical level. Formations with bold are used in reconstructions, (rot.) = the pole is rotated according to an Euler pole (51,275,−14) by Heaman (2004), Plat, Plong = latitude and longitude of the paleomagnetic pole, A95 = 95% confidence cone around the pole, Age of ChRM (Ma) = age of the characteristic remanent magnetization. References: 1 = This study, 2 = Bibikova et al. (2005), (3) = Halla (2002), (4) = Mertanen et al. (2006a), (5) = Mänttäri and Hölttä (2002), (6) = Neuvonen et al. (1997), (7) = Mertanen et al. (2006b), (8) = Bleeker et al. (2008), (9) = Krasnova and Gooskova (1990), (10) = Sergeev et al. (1990), (11) = Mertanen et al. (2006b), (12) = Mertanen et al. (1999), (13) = Pullaiah and Irving (1975), (14) = Buchan et al. (1990), (15) = Zhai et al. (1994), (16) = Heaman et al. (1986), (17) = Fahrig et al. (1986), (18) = Buchan et al. (1998), (19) = Evans and Halls (2010), (20) = Heaman (2004).

terozoic remanences. In the Vodlozero terrane the Archean and Early Paleoproterozoic rocks do not show any evidence of Svecofennian overprinting, suggesting that reworking of the crust associated with Svecofennian accretion did not significantly affect the Vodlozero terrane. In the following sections, the paleomagnetic data from these formations are compared with the results from the Koitere granitoids. The paleomagnetic poles from the Koitere, Varpaisjärvi and Vodlozero areas are shown in Table 2 and on Fig. 10.

5.2.1. Varpaisjärvi

The paleopole from the Varpaisjärvi area (Fig. 1) is based on the mean value of 14 sites comprising enderbites and mafic granulites (Mertanen et al., 2006a). The pole previously obtained by Neuvonen et al. (1981, 1997) from the Varpaisjärvi area is in agreement with this pole (Table 2). U–Pb zircon ages of the enderbites have a range of ca. 2690–2680 Ma (Paavola, 1984, 1986Paavola (1988)), and the U–Pb ages of metamorphic zircons and monazites are ca. 2630 Ma (Hölttä et al., 2000). According to Mänttäri and Hölttä (2002), the ion microprobe U–Pb results show ages of ca. 2720–2700 Ma from magmatically zoned zircons, and ages of ca. 2650–2640 Ma from metamorphic zircon overgrowths, suggesting that the conventional U–Pb zircon ages between 2690 and 2680 Ma represent mixing between two zircon age components. Furthermore, the Sm–Nd garnet–whole rock ages of ca. 2590–2480 Ma are inferred to record temperatures of ~600 °C associated with cooling from conditions of granulite-grade metamorphism (Hölttä et al., 2000). As discussed in Mertanen et al. (2006a), the Sm–Nd garnet ages likely record the timing of remanent magnetization as the Sm–Nd closure temperature may be similar to the blocking temperature of magnetite. Therefore, the age of ca. 2680 Ma for the Varpaisjärvi remanence used in some recent comparisons between different Archean cratons (Evans and Pisarevsky, 2008) is likely too old.

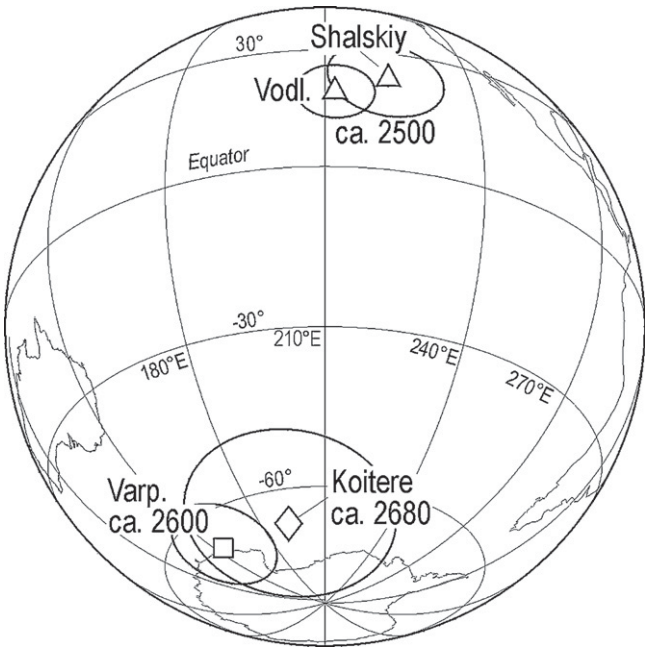


Fig. 10. Archean paleomagnetic poles from the Koitere sanukitoids (Koitere), Varpaisjärvi enderbites and granulites (Varp.), Shalski gabbronorite dyke (Shalskiy) and Vodla River gneisses (Vodl.). The open symbols denote reversed polarity.

5.2.1.1. Comparison of paleomagnetic results from Koitere and Varpaisjärvi. The Varpaisjärvi enderbites show a steep characteristic remanence component, similar to the results from Koitere, but with a positive downward-pointing inclination (Mertanen et al., 2006a). This difference may suggest that the Koitere granitoids and the Varpaisjärvi enderbites record a long-lived Neoarchean metamorphic event during which the polarity of the Earth's magnetic field reversed at least once. Metamorphism took place during the amalgamation of Archean crust in several successive collisions According to Lobach-Zhuchenko et al. (2000), the West and Central Karelian terranes and the Vodlozero terrane had formed an integrated craton by ca. 2.74–2.72 Ga and the crustal blocks did not experience additional relative movement, whereas Käpyaho (2007 and references therein) proposed that the crustal domains were juxtaposed by ca. 2.67 Ga to form a coherent craton. Kontinen et al. (2007) suggested that high-grade metamorphism was a craton-wide event between ca. 2.66 and 2.63 Ga, with the possibility that elevated temperatures may have been sustained during the interval of ca. 2.70–2.63 Ga. Hölttä et al. (2000) proposed that in the

Varpaisjärvi area the peak of metamorphism took place at ca. 2.63 Ga and may have lasted up to ca. 2.60 Ga during the final amalgamation with the eastern Karelia craton. Despite the slight differences between these models, the geological evidence suggests that the Archean crustal blocks of Fennoscandia had amalgamated by ca. 2.6 Ga and leaves open the possibility of protracted metamorphism.

The steep remanence directions with opposite polarities from the Koitere and Varpaisjärvi areas are statistically undistinguishable (Table 2, Fig. 10), although the exact magnetization ages associated with cooling from peak metamorphic conditions are uncertain in both formations. However, it is likely that they record Archean metamorphic events at slightly different times: the Koitere charno–enderbites in eastern Finland at ca. 2.68–2.63 Ga ($^{207}Pb/^{206}Pb$ monazite age of 2684 ± 2 Ma, Halla (2002)), and the Varpaisjärvi granulites in western Finland slightly later, at ca. 2.63–2.60 Ga (U–Pb zircon and monazite ages of ca. 2630 Ma, Hölttä et al., 2000). Within error limits, the Koitere paleopole positions the craton at a high paleolatitude of 85–70° at ca. 2.68–2.63 Ma (Fig. 11a), and the paleopole from Varpaisjärvi positions the craton at a paleolatitude of 70–60° between ca. 2.63 and 2.60 Ma (Fig. 11b). Taking into account the uncertainties in the magnetization ages and the circles of confidence of the poles, the combined data from Koitere and Varpaisjärvi are consistent with high paleolatitudes for the Karelia craton at ca. 2.7–2.6 Ga.

5.2.2. Vodlozero terrane and Shalskiy gabbronorite dyke

The Vodlozero terrane in the most SE part of the Karelia craton (Fig. 1) forms part of the oldest Archean basement in the Fennoscandian shield, with protolith ages of the gneisses as old as ca. 3.2 Ga (Sergeev et al., 1990). A paleomagnetic study on the Vodlozero terrane and the Shalskiy gabbronorite dyke was carried out previously using a Sm–Nd model age of 2608 ± 56 Ma (Mertanen et al., 2006b). However, recent U–Pb baddeleyite ages of 2510 ± 1.6 Ma suggest that the dyke is clearly younger (Bleeker et al., 2008), therefore the data and tectonic implications are discussed here. The pole from the Shalskiy gabbronorite dyke differs significantly from the poles of Varpaisjärvi and Koitere (Fig. 10). The Shalskiy dyke has a shallow southward, upward- or downward-pointing direction (Krasnova and Gooskova, 1990; Mertanen et al., 2006b). This shallow inclination component is interpreted as the ca. 2.50 Ga remanence of the dyke. However, because this pole was obtained only from one dyke, it is a virtual geomagnetic pole (VGP) and cannot alone represent the ca. 2.50 Ga magnetization. A similar low inclination remanence direction was also obtained in the host basement gneisses and in the Vodla River gneisses (Krasnova and Gooskova, 1990) located ca. 50 km east from the Shalskiy dyke (Table 2, Fig. 10). Accordingly, the results from the dyke fail a baked contact test. It is therefore possible that emplacement of the Shalskiy dyke and subsequent magnetization were associated with reheating or metasomatism in the Vodlozero domain at ca. 2.50 Ga. U–Pb apatite ages of 2540 ± 40 Ma have been interpreted by Sergeev et al. (1990) to constrain the latest timing of Archean metasomatic reworking of the Vodlozero terrane, and according to Ovchinnikova et al. (1991), late kinematic granites experienced metasomatism as late as 2500 ± 10 Ma, providing further evidence for Neoarchean reworking in the Vodlozero area. Consequently, it cannot be ruled out that the low inclination component of the Shalskiy dyke and the Vodlozero basement rocks represents a 2.50 Ga remagnetization of the area.

Consistent with the U–Pb age of the Shalskiy dyke (2.50 Ga) and the Sm–Nd model age (2.59 Ga) from the Varpaisjärvi granulites, there is at least a 90 Ma age difference between the magnetizations. The data also suggest substantial plate movement during this time interval (Fig. 11). Based on the Koitere and Varpaisjärvi data, the craton occupied high paleolatitudes of about 85–60° at ca. 2·7–2·6 Ma (Fig. 11a and b). At that time the craton cooled from conditions of high-grade metamorphism. By ca. 2.50 Ga, it had drifted to an equatorial paleolatitude when rifting and mild reworking took place in the eastern part of the craton (Fig. 11c).

5.3. Reconstructions with the Superior craton

Cratonisation and granitoid magmatism at ca. 2.70–2.63 Ga was a worldwide phenomenon (e.g. Bleeker, 2003; Condie et al., 2009). Neoarchean-early Paleoproterozoic development of the Superior craton of Laurentia has been correlated with events in the Fennoscandian shield, as they share several common geological features. In order to make continental reconstructions between the Superior and Karelia cratons, we use Neoarchean paleomagnetic data from the Superior craton listed in the Global Paleomagnetic Database (GPDB) and in recent publications.

The Archean Superior craton provides several well-defined paleomagnetic poles, but according to Evans and Pisarevsky (2008) only a few of the data satisfy the most stringent reliability criteria. In many cases the primary nature of the remanent magnetization has not been demonstrated by field tests, and therefore the magnetization ages for many poles are not well constrained. In the following, three poles from the Superior craton with ages of ca. 2680 Ma, 2640 Ma and 2505 Ma are compared with similar-aged data from Fennoscandia. All poles are comparatively well-defined and the authors have evidence to show that the magnetization ages are in accord with isotopic ages. Although the poles are not unequivocally primary, they are used here cautiously to compare with the Fennoscandian data.

The oldest pole (Table 2) is from the Otto stock lamprophyre dykes and aureole (OS, N + R component, Pullaiah and Irving, 1975) from the Abitibi Belt in the southern Superior craton, dated at 2680 ± 1 Ma (U–Pb zircon, Corfu et al., 1989). The pole has been rotated to the eastern reference frame of Superior according to the Euler pole (51°N, 275°E, −14°) provided by Evans and Halls (2010). The remanence direction of the Otto stock dykes is well-defined (Buchan et al., 1990), and the pole places the Superior craton at a high paleolatitude of about 60°. The age of the Otto stock corresponds with the metamorphic age of the Koitere granitoids (2684 ± 2 Ma), and they are used here as a pair for a ca. 2.68–2.63 Ga reconstruction (Fig. 11a). According to this reconstruction, both cratons occupied high paleolatitudes at that time. In the Karelia craton, the ca. 2.74–2.72 Ga sanukitoids have an approximate N–S trend, whereas in the Superior craton, the ca. 2.72–2.68 Ga sanukitoids were emplaced in an approximate E–W belt (J. Halla, pers. comm., 2007). The reconstruction in Fig. 11a is for a younger time than the emplacement age of the sanukitoids, and as such cannot be used to constrain whether the sanukitoids were emplaced on a united craton at ca. 2.72 Ga. In this reconstruction, the cratons are placed as close to each other as possible within the error limit (A95 = 19.5°) of the Koitere pole. Accordingly, the reconstruction could allow the cratons to have been joined, provided that the Karelia craton was located on the western side of the present Superior craton. However, the ca. 2.6 Ga and 2.5 Ga poles for Superior and Karelia, consistent with this configuration (Fig. 11a), shows that this reconstruction did not last up to those times.

The second pole from the Superior craton is from the Pickwitonei Granulite (PG, D component in Zhai et al., 1994) in the northwestern part of the craton (Table 2). The Pickwitonei pole is rotated to the eastern reference frame according to Evans and Halls (2010). The pole is well-defined although the magnetization age has not been constrained by field tests. The emplacement and metamorphic evolution of the Pickwitonei granulite shows similarities with the Varpaisjärvi granulites. U–Pb zircon ages of the Pickwitonei granulite are 2695 ± 2 Ma and 2637 ± 2 Ma (Heaman et al., 1986), and are similar to U–Pb zircon ages from

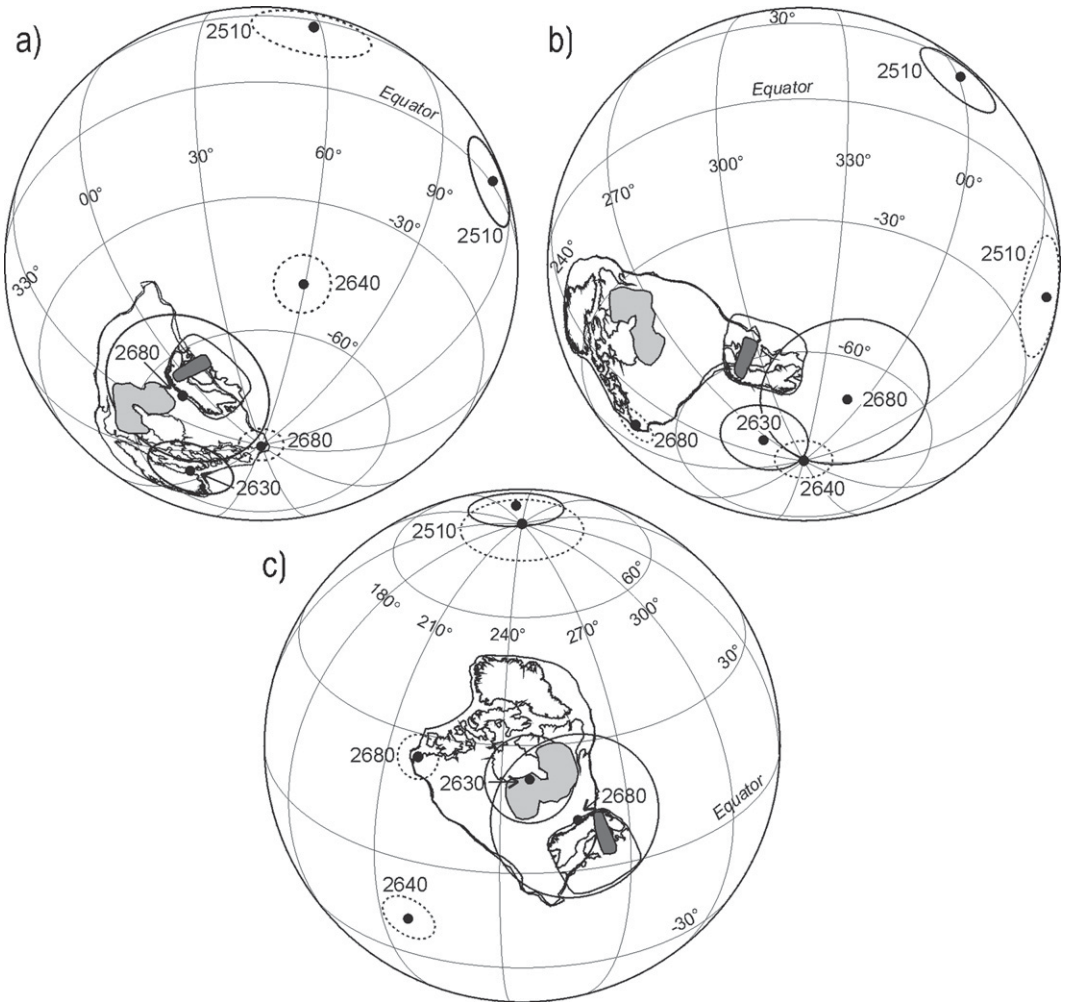


Fig. 11. Reconstructions of Superior and Karelia in Neoarchean at ca. 2.7–2.5 Ga. (a) Reconstruction based on poles from the Koitere granitoids (maximum age 2684 ± 2 Ma) and Otto stock lamprophyre (2680 ± 1 Ma). Euler parameters (latitude, longitude, angle) for the rotation of Karelia and its poles (5.7, 250.0, −139.5) and for Superior (0, 137, −159), (b) Reconstruction based on poles from the Varpaisjärvi enderbites and granulites (ca. 2630–2480 Ma) and the Pickwitonei granulite (ca. 2640–2576 Ma). Euler parameters for Karelia and its poles (10.7, 195, −155.2) and for Superior (0, 127.9, −117.9), (c) Reconstruction based on poles from the Shalskiy dyke (2510 ± 1.6 Ma) and Ptarmigan–Mistassini dykes (ca. 2515–2503 Ma), Euler parameters for Karelia and its poles (59, 247.5, −141) and for Superior (0, 303, 44.7). Poles of Karelia are shown as solid-line circles and of Superior as dotted-line circles. The ages (Ma) give the approximate timing for remanence acquisition. In all figures Superior is rotated to its absolute framework, and is represented by light gray shading. Karelia is rotated to Superior by taking into account the confidence limits of the poles, and is represented by dark gray shading.

the Varpaisjärvi area. The U–Pb garnet age of 2640 Ma records the peak of the Kenoran granulite-grade metamorphism (Mezger et al., 1990), and is nearly contemporaneous with the peak metamorphism in the Varpaisjärvi area. The Sm–Nd garnet–whole rock ages of 2605 ± 9 Ma and 2576 ± 5 Ma from the Pickwitonei granulites record cooling through temperatures of 650–600 °C (Mezger et al., 1992), similar to results from Varpaisjärvi. A recent study of U–Pb ages of rutile using laser ablation ICP-MS (Kooijman et al., 2010) shows that the ages are as young as ca. 2.3 Ga, significantly younger than the crystallization age of the prograde to peak metamorphic assemblage (ca. 2640 Ma). In the Varpaisjärvi area, laser ablation ICP-MS ages are not available, but due to similarities obtained from other chronometric methods and the resemblance in metamorphic cooling history, as well as the similarity between the well-defined paleomagnetic poles of the Pickwitonei and Varpaisjärvi granulites, these two are taken as one pair for reconstruction (Fig. 11b). Taking the youngest age estimates for cooling below the magnetite blocking temperature, we conclude that at ca. 2640–2576 Ma the Superior craton was located at a paleolatitude of about 40°, and at ca. 2630–2480 Ma the Karelia craton was located at 70–60°. The reconstruction suggests that Karelia was located on the western side of Superior, and that the two cratons were separated at those times. Furthermore, the discrepancy between the Superior and Karelia poles at ca. 2.68 Ma and ca. 2.50 Ma (Fig. 11b) shows that the cratons were probably still moving with respect to each other. These results contradict proposed supercontinental models that position Karelia south of Superior and invoke a direct connection between them, with amalgamation between ca. 2.72 Ga and 2.68 Ga (e.g. Bleeker and Ernst, 2006 and references therein). Both cratons do show some drift from higher to lower paleolatitudes between ca. 2.68–2.60 Ga, which may indicate that they were part of a larger supercontinent, e.g. Kenorland (Aspler and Chiarenzelli, 1998; Hoffman, 1992). The primary nature of the remanence has not been demonstrated by proper field tests in either the Pickwitonei granulites or the Varpaisjärvi granulites, which casts doubt on the ages of remanence. Therefore, the main conclusion from the present paleomagnetic data is that the Superior and Karelia cratons were in close proximity at ca. 2.68–2.60 Ga, but the data cannot demonstrate that the cratons were contiguous at that time.

The third Archean pole from the Superior craton (Table 2) is from the Ptarmigan dyke swarm in the northeastern part of the craton, dated at 2505 ± 2 Ma (Fahrig et al., 1986; Buchan et al.,

1998). The Ptarmigan dykes have similar age and paleomagnetic record as the recently dated ca. 2510–2515 Ma (Heaman, 2004) Mistassini dykes (Table 2) in the eastern Superior craton (Buchan et al., 2007). The combined pole from the Ptarmigan and Mistassini dykes (Evans and Halls, 2010) is used to compare with data from the ca. 2510 Ma Shalskiy dyke in Karelia. Uncertainties on the ages of the Ptarmigan–Mistassini and Shalskiy poles arise from the fact that the remanence of neither of the dyke swarms have been proven to be primary by strict field tests (Buchan et al., 2007; Mertanen et al., 2006b). The reconstruction for that time is shown in Fig. 11c. Compared to the previous configuration (Fig. 11b), the Superior and Karelia cratons have very different relative paleopositions, although both cratons are located near the equator (Fig. 11c). The Superior craton has drifted across the equator to a latitude of about 20° and rotated clockwise about 45°. At the same time, the Karelia craton drifted about 70° to the equator and rotated about 120° clockwise. This reconstruction is in close agreement with the Superia model of Bleeker and Ernst (2006), in which the 'greater' Karelia (Karelia + Kola), and the Hearne and Wyoming cratons are located along the southern margin of the Superior craton. The model is based on several Archean–Paleoproterozoic geological similarities (matching barcodes) (Bleeker, 2003, 2004; Bleeker and Ernst, 2006) and orientations of similar-aged dyke swarms on the cratons. The 2.5 Ga Ptarmigan, Mistassini and Shalskiy dykes are believed to radiate from a common plume center (Bleeker and Ernst, 2006). However, if the Shalskiy dyke represents one part of the plume-related radial dyke system, its present NE–SW orientation is not in accord with the model, which requires a more NW–SE orientation (Bleeker and Ernst, 2006). In that sense the present paleomagnetic data cannot confirm the presented model for 2.50 Ga, although the proximity of the cratons would support it.

6. Conclusions

The results from this study suggest that a ca. 2.68–2.63 Ga Neoarchean remanence with steep upward-pointing inclination has been preserved in the Koitere charno-enderbitic granitoids. Stable and consistent results were obtained from 12 sites in areas with high positive magnetic anomalies and high remanence intensities, whereas the Archean TTG gneisses and sheared rocks surrounding the granitoids carry a recent viscous remanence or an overprint acquired during the ca. 1.9–1.8 Ga Svecofennian orogeny. Electron microprobe and petrographic observations coupled with rock magnetic studies show that the ChRM resides in fine-grained SD/PSD magnetite possibly formed during Neoarchean hydrous alteration of clinopyroxene. Based on the occurrence of cross-cutting vertical/subvertical Paleoproterozoic ca. 2.5–1.95 Ga dolerite dykes, it is implied that the Koitere granitoids record paleohorizontal orientations and that the remanence direction can be used for paleoreconstructions.

The steep negative inclination remanence direction from the Koitere granitoids is opposite in polarity to the ca. 2.6 Ga Varpaisjärvi enderbites and granulites, suggesting a polarity reversal occurred during the Neoarchean. The data from Koitere and Varpaisjärvi imply that at ca. 2.7–2.6 Ga the Karelia craton was located at high paleolatitudes of 80–60°. Previous paleomagnetic data from 2.5 Ga formations in the Vodlozero terrane position Karelia at a near-equatorial latitude at ca. 2.50 Ga.

Comparison of paleomagnetic data from the Koitere and Varpaisjärvi granulite-grade rocks with data from rocks of similar age in Superior (2680 Ma Otto stock lamprophyre dykes and aureole, and 2640 Ma Pickwitonei granulite) shows that at ca. 2.7–2.6 Ga the Superior and Karelia cratons were in close proximity to each other. However, the present paleomagnetic data cannot demonstrate that the cratons were joined at that time.

During the Archean–Paleoproterozoic transition at 2.50 Ga both the Karelia and Superior cratons experienced significant rotation and drifting: Superior to near-equatorial paleolatitude and Karelia to the equator, thus suggesting that they may have been together at that time.

Acknowledgements

Asko Kontinen and Peter Sorjonen-Ward provided geological expertise both in discussions and in the field and showed the locations for sampling. Both are greatly acknowledged. Meri-Liisa Airo is thanked for aeromagnetic data and interpretations. Lauri Pesonen, Arto Kataikko, Sami Lähde, Keijo Kinnunen and Carmen Eyssautier took part in the field sampling and are thanked for their essential work and companionship. Matti Leino and Tuula Laine are thanked for their work in the paleomagnetic laboratory. Esa Heilimo and Jaana Halla are acknowledged for discussions and comments on the manuscript. Reviews by Sergei Pisarevsky and Mike Wingate helped significantly to improve the paper. A Fulbright Fellowship allowed Korhonen to conduct some of this work at the Geological Survey of Finland, and the U.S Department of State is acknowledged for funding support.

References

Aspler, L.B., Chiarenzelli, J.R., 1998. Two Neoarchean supercontinents? Evidence from the Paleoproterozoic. Sedimen. Geol. 120, 75–104.
Bibikova, E.V., Petrova, A., Claesson, S., 2005. The temporal evolution of sanukitoids in the Karelian Craton, Baltic Shield: an ion microprobe U–Th–Pb isotopic study of zircons. Lithos 79, 129–145.
Bleeker, W., 2003. The late Archean record: a puzzle in ca. 35 pieces. Lithos 71, 99–134.
Bleeker, W., 2004. Taking the pulse of planet earth: a proposal for a new multi-disciplinary flagship project in Canadian solid earth sciences. Geosci. Canada 31, 179–190.
Bleeker, W., Ernst, R., 2006. Short-lived mantle generated magmatic events and their dyke swarms: the key unlocking Earth's paleogeographic record back to 2.6 Ga. In: Hanski, E., Mertanen, S., Rämö, T., Vuollo, J. (Eds.), Dyke Swarms—Time Markers of Crustal Evolution. Taylor & Francis Group, London, pp. 3–26.
Bleeker, W., Hamilton, M.A. Ernst, R.E., Kulikov, V.S. 2008. The search for Archean-Paleoproterozoic supercratons; new constraints on Superior-Karelia-Kola correlations within supercraton Superia, including the first ca. 2504 Ma (Mistassini) ages from Karelia. 33rd International Geological Congress, Abstracts.
Buchan, K.L., Neilson, D.J., Hale, C.J., 1990. Relative age of Otto stock and Matachewan dykes from paleomagnetism and implications for the Precambrian polar wander path. Can. J. Earth Sci. 27, 915–922.
Buchan, K.L., Mortensen, J.K., Card, K.D., Percival, J.A., 1998. Paleomagnetism and U–Pb geochronology of diabase dyke swarms of Minto block, Superior Province, Quebec, Canada. Can. J. Earth Sci. 35, 1054–1069.
Buchan, K.L., Goutier, J., Hamilton, M.A., Ernst, R.E., Matthews, W.A., 2007. Paleomagnetism, U–Pb geochronology, and geochemistry of Lac Esprit and other dyke swarms, James Bay area, Quebec, and implications for Paleoproterozoic deformation of the Superior Province. Can. J. Earth Sci. 44, 643–664.
Condie, K.C., Belousova, E., Griffin, W.L., Sircombe, K.N., 2009. Granitoid events in space and time: constraints from igneous and detrital zircon age spectra. Gondwana Res. 15, 228–242.
Corfu, F., Krogh, T.E., Kwok, Y.Y., Jensen, L.S., 1989. U–Pb zircon geochronology in the southwestern Abitibi greenstone belt, Superior Province. Can. J. Earth Sci. 26, 1747–1763.
de Kock, M.O., Evans, D.A.D., Beukes, N.J., 2009. Validating the existence of Vaalbara in the Neoarchean. Precamb. Res. 174, 145–154.
Dunlop, D.J., 1983. Determination of domain structure in igneous rocks by alternating field and other methods. Earth Planet. Int. 26, 1–26.
Evans, D.A.D., Halls, H.C., 2010. Restoring Proterozoic deformation within the Superior craton. Precamb. Res. 183, 474–489.
Evans, D.A.D., Pisarevsky, S.A., 2008. Plate tectonics on early Earth? Weighing the paleomagnetic evidence. In: Condie, K.C., Pease, V. (Eds.), When Did Plate Tectonics Begin on Planet Earth? Geol. Soc. America Spec. Paper 440, p. 249–263.
Fahrig, W.F., Christie, K.W., Chown, E.H., Janes, D., Machado, N., 1986. The tectonic significance of some basic dyke swarms in the Canadian Superior Province with special reference to the geochemistry and paleomagnetism of the Mistassini swarm, Quebec, Canada. Can. J. Earth Sci. 23, 238–253.
Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. London, Serie A 217, 293–305.
Halla, J., 2002. Origin and Paleoproterozoic reactivation of Neoarchean high-K granitoid rocks in eastern Finland. PhD Thesis, University of Helsinki, Finland. Annale Academica Scientiarium Fennicae, Geologica–Geographica 163, 105 pp.

Halla, J., 2005. Late Archean high-Mg granitoids (sanukitoids) in the southern Karelian domain, eestern Finland: Pb and Nd isotopic constraints on crust–mantle interactions. Lithos 79, 161–178.
Halla, J., van Hunen, J., Heilimo, E., Hölttä, P., 2009. Geochemical and numerical constraints on Neoarchean plate tectonics. Precamb. Res. 174, 155–162.
Heaman, L.M., 2004. 2.5–2.4 Ga global magmatism: remnants of supercontinents or products of superplumes. Geological Society of America, Denver Conference, November 2004, Abstracts 36, 255.
Heaman, L.M., Machado, N., Krogh, T.E., Weber, W., 1986. Preliminary U-Pb zircon results from the Pickwitonei granulite domain, Manitoba. In: Geological Association of Canada–Mineralogical Association of Canada Joint Annual Meeting 1986, Ottawa, Ontario, Program and Abstracts, 79.
Heilimo, E., Mikkola, P., Halla, J., 2007. Age and petrology of the Kaapinsalmi sanukitoid intrusion in Suomussalmi, eastern Finland. Bull. Geol. Soc. Finland 79, 117–125.
Heilimo, E., Halla, J., Hölttä, P., 2010a. Discrimination and origin of the sanukitoid series: geochemical constraints from the Neoarchean western Karelian Province (Finland). Lithos 115, 27–39.
Heilimo, E., Halla, J., Huhma, H., 2011. Single-grain zircon U–Pb age constraints of the western and eastern sanukitoid zones in the Finnish part of the Karelian Province. Lithos 121, 87–99.
Hoffman, P.F., 1992. Supercontinents. Encyclopedia of Earth System Science, 4. Academic Press, London, pp. 323–328.
Hölttä, P., Huhma, H., Mänttäri, I., Paavola, J., 2000. P-T-t development of Archaean granulites in Varpaisjärvi, Central Finland, II: dating of high-grade metamorphism with the U–Pb and Sm–Nd methods. Lithos 50, 121–136.
Hölttä, P., Balagansky, V., Garde, A.A., Mertanen, S., Peltonen, P., Slabunov, A., Sorjonen-Ward, P., Whitehouse, M., 2008. Archean of Greenland and Fennoscandia. Episodes 31 (1), 13–19.
Kelsey, D.E., Clark, C., Hand, M., 2008. Thermobarometric modelling of zircon and monazite growth in melt-bearing systems: examples using model metapelitic and metapsammitic granulites. J. Metamorph. Geol. 26, 199–212.
Käpyaho, A., 2007. Archaean crustal evolution in eastern Finland: new geochronological and geochemical constraints from the Kuhmo terrain and the Nurmes belt. PhD Thesis, Department of Geology, University of Helsinki. Geological Survey of Finland, 20, 97 pp. + 1 app.
Käpyaho, A., Hölttä, P., Whitehouse, M.J., 2007. U–Pb zircon geochronology of selected Archaean migmatites in eastern Finland. Bull. Geol. Soc. Finland 79, 95–115.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geoph. J. R. Astr. Soc. 62, 699–718.
Koistinen, T., Stephens, M.B., Bogatchev, V., Nordgulen, Ø., Wennerström, M., Korhonen, J. (comps.) 2001. Geological map of the Fennoscandian Shield, scale 1:2 000 000. Geol. Surv. Finland, Geol. Surv. Norway, Geol. Surv. Sweden, Ministry of Natural Resources of Russia.
Kooijman, E., Mezger, K., Berndt, J., 2010. Constraints on the U–Pb systematics of metamorphic rutile from in situ LA-ICP-MS analysis. Earth Planet. Sci. Lett. 293, 321–330.
Kontinen, A., Paavola, J., 2006. A preliminary model of the crustal structure of the eastern Finland Archaean complex between Vartius and Vieremä, based on constraints from surface geology and FIRE 1 seismic survey. In: Finnish Reflection Experiment FIRE 2001–2005. Geol. Surv. Finland. Spec. Paper 43, 223–240.
Kontinen, A., Käpyaho, A., Huhma, H., Karhu, J., Matukov, D.I., Larionov, A., Sergeev, S.A., 2007. Nurmes paragneisses in eastern Finland, Karelian craton: provenance, tectonic setting and implications for Neoarchaean craton correlation. Precamb. Res. 152, 119–148.
Korhonen, F., Sorjonen-Ward, P., Mertanen, S., Airo, M.-L., 2006. Magnetic signatures of Archean and Proterozoic magmatic and metamorphic events in the Lieksa area, eastern Finland. Bull. Geol. Soc. Finland 1 (special issue), 72.
Krasnova, A.F., Gooskova, E.G., 1990. Geodynamic evolution of the Wodlozero block of Karelia according to palaeomagnetic data. Earth Physics, Izvestiya 26, 80–85.
Krasnova, A.F., Gooskova, E.G., 1995. Palaeomagnetism of Precambrian basic intrusion and dykes of Northern Karelia, eastern Fennoscandian Shield. Precamb. Res. 74, 245–252.
Leino, M.A.H., 1991. Paleomagneettisten tulosten monikomponenttianalyysi pienimmän neliösumman menetelmällä. (Multicomponent analysis of palaeomagnetic data by least square method). Lab. Palaeomagn., Dept. Geoph., Geol. Surv. Finland, Rep. Q29.1/91/2, 15 pp. (in Finnish).
Lobach-Zhuchenko, S.B., Chekulaev, V.P., Arestova, N.A., Levskii, L.K., Kovalenko, A.V., 2000. Archean terranes in Karelia: geological and isotopic-geochemical evidence. Geotectonics 34, 452–466.
Lobach-Zhuchenko, S.B., Rollinson, H.R., Chekulaev, V.P., Arestova, N.A., Kovalenko, A.V., Ivanikov, V.V., Guseva, N.S., Sergeev, S.A., Matukov, D.I., Jarvis, K.E., 2005. The Archean sanukitoid series of the Baltic Shield: geological setting, geochemical characteristics and implications for their origin. Lithos 79, 107–128.
Lowrie, W., Fuller, M., 1971. On the alternating field demagnetization characteristics of multidomain thermoremanent magnetization in magnetite. J. Geoph. Res. 76, 6339–6349.
Mänttäri, I., Hölttä, P., 2002. U–Pb dating of zircons and monazites from Archean granulites in Varpaisjärvi, central Finland: evidence for multiple metamorphism and Neoarchean terrane accretion. Precamb. Res. 118, 101–131.
Mertanen, S., 1995. Multicomponent remanent magnetizations reflecting the geological evolution of the Fennoscandian Shield – a palaeomagnetic study with emphasis on the Svecofennian orogeny. PhD Thesis, Department of Geology, University of Helsinki. Geological Survey of Finland, 46 pp., with original articles I–IV.
Mertanen, S., Korhonen, F., 2008. Archean-Paleoproterozoic configuration of Laurentia and Baltica focusing on paleomagnetic data from Baltica. 33rd International Geological Congress, 6–14 August 2008, Oslo, Norway. Abstracts.
Mertanen, S., Pesonen, L.J., Huhma, H., Leino, M.A.H., 1989. Palaeomagnetism of the Early Proterozoic layered intrusions, northern Finland. Geol. Surv. Finland, Bull. 347, 40 pp.
Mertanen, S., Pesonen, L.J., Huhma, H., 1996. Palaeomagnetism and Sm-Nd ages of the Neoproterozoic diabase dykes in Laanila and Kautokeino, BT northern Fennoscandia. In: Brewer, T. (Ed.), Precambrian Crustal Evolution in the North Atlantic Region, 112. Geological Society Special Publication, pp. 331–358.
Mertanen, S., Halls, H.C., Vuollo, J., Pesonen, L.J., Stepanov, V.S., 1999. Paleomagnetism of 2.44 Ga mafic dykes in Russian Karelia, eastern Fennoscandian shield – implications for continental reconstructions. Precamb. Res. 98, 197–221.
Mertanen, S., Pesonen, L.J., Hölttä, P., Paavola, J., 2006a. Palaeomagnetism of Palaeoproterozoic dolerite dykes in central Finland. In: Hanski, E., Mertanen, S., Rämö, O.T., Vuollo, J. (Eds.), Dyke Swarms—Time Markers of Crustal Evolution. Taylor & Francis Group, London, pp. 243–256.
Mertanen, S., Vuollo, J.I., Huhma, H., Arestova, N.A., Kovalenko, A., 2006b. Early Paleoproterozoic-Archean dykes and gneisses in Russian Karelia of the Fennoscandian Shield–new paleomagnetic, isotope age and geochemical investigations. Precamb. Res. 144 (3–4), 239–260.
Mezger, K., Bohlen, S.R., Hanson, G.N., 1990. Metamorphic history of the Archean Pikwitonei granulite domain and the Cross Lake Subprovince, Superior Province, Manitoba, Canada. J. Petrol. 31, 483–517.
Mezger, K., Essene, E.J., Halliday, A.N., 1992. Closure temperatures of the Sm–Nd system in metamorphic garnets. Earth Planet. Sci. Lett. 113, 397–409.
Nehring, F., 2007. Partial melting within the lower crust–Constraints from bulk rock geochemical data and trace element mineral analyses of granulites and migmatites from Central Finland. PhD Thesis, Johannes Gutenberg-University Mainz, Institute of Geosciences.
Neuvonen, K.J., Korsman, K., Kouvo, O., Paavola, J., 1981. Paleomagnetism and age relationship of the rocks in the Main Sulphide Ore Belt in central Finland. Bull. Geol. Soc. Finland 53, 109–133.
Neuvonen, K.J., Pesonen, L.J., Pietarinen, H., 1997. Remanent Magnetization in the Archaean basement and cutting diabase Dykes in Finland, Fennoscandian Shield. Geophysica 33, 111–146.
Ovchinnikova, G.V., Lobach-Zhuchenko, S.B., Sergeyev, S.Z., Yakovleva, S.Z., Levchenkov, O.A., Neymark, L.A., Komarov, A.N., Gorokhovskiy, B.M., Fedoseynko, A.M., Krylov, I.N., 1991. Geochemical and isotope data on the dating and petrology of southeast Karelian late-kinematic granites. Geochem. Int. 28, 35–47.
Paavola, J., 1984. On the Archaean high-grade metamorphic rocks in the Varpaisjärvi area, Central Finland. Geological Survey of Finland, Bulletin 327, 33 pp.
Paavola, J. 1988. Lapinlahden alueen kallioperä. Summary: Pre-Quaternary rocks of the Lapinlahti Map Sheet area. Geological Map of Finland 1:100,000, Explanation to the maps of Pre-Quaternay Rocks, Sheet 3332. Geological Survey of Finland.
Pullaiah, G., Irving, E., 1975. Paleomagnetism of the contact aureole and late dikes of the Otto Stock, Ontario, and its application to Early Proterozoic apparent polar wandering. Can. J. Earth Sci. 12, 1609–1618.
Sergeev, S.A., Bibikova, E.V., Levchenkov, O.A., Lobach-Zchuchenko, S.B., Yakoleva, S.Z., Ovchinnikova, G.V., Neymark, L.A., Komarov, A.N., Gorokhovsky, B.M., 1990. Isotope geochronology of the Vodlozero gneiss complex. Geochem. Int. 27, 65–74.
Slabunov, A.I., Lobach-Zhuchenko, S.B., Bibikova, E.V., Sorjonen-Ward, P., Balagansky, V.V., Volodichev, O.I., Shcipansky, A.A., Svetov, S.A., Chekulaev, V.P., Arestova, N.A., Stepanov, V.S., 2006. The Archean nucleus of the Fennoscandian (Baltic) shield. In: Gee, D.G., Stephenson, R.A. (Eds.), European Litospehere Dynamics. Memoirs, Geol. Soc. London 32, pp. 627–644.
Strik, G., Blake, T.S., Zegers, T.E., White, S.H., Langereis, C.G., 2003. Paleomagnetism of flood basalts in the Pilbara Craton, western Australia: late Archean continental drift and the oldest known reversal of the geomagnetic field. J. Geophys. Res. 108, B12, 2551, EPM 2, 1–21.
Strik, G., de Wit, M.J., Langereis, C.G., 2007. Paleomagnetism of Neoarchean Pongola and Ventersdorp Supergroups and an apprisal of the 3.0–1.9 Ga apparent polar wander path of the Kaapvaal Craton, Southern Africa. Precamb. Res. 153, 96–115.
Vaasjoki, M., Taipale, K., Tuokko, I., 1999. Radiometric ages and other isotopic data bearing on the evolution of Archaean crust and ores in the Kuhmo-Suomussalmi area, eastern Finland. Studies Related to the Global Geoscience Transects/SVEKA Project in Finland. Bull. Geol. Soc. Finland 71, 155–176.
Vuollo, J., Huhma, H., 2005. Paleoproterozoic mafic dykes in NE Finland. In: Lehtinen, M., Nurmi, P.A., Rämö, O.T. (Eds.), Precambrian Geology of Finland–Key to the Evolution of the Fennoscandian Shield. Elsevier Science B.V., Amsterdam, 193-235.
Zhai, Y., Halls, H.C., Bates, M.P., 1994. Multiple episodes of dike emplacement along the northwestern margin of the Superior Province, Manitoba. J. Geophys.Res. 99, 21717–21732.
Zijderveld, J.D., Huhma, H., 1967. A.C. demagnetization in rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, New York, 254-286.